

(CART) Analysis Project Write Up Summary

Where the Tree Learns:

The most important thing the tree notices is shelf location. The very first split the tree makes is whether a store has "Good" shelf location or not. If a store has good shelf location, the tree expects much higher sales (about 10k units) compared to stores with bad or medium shelf location (about 6.8k units). Of course, this makes sense: if your product is right in front of people's faces versus stuck in a corner, more people are obviously going to buy it.

After this first split, the tree starts paying a lot of attention to price. Almost every split, it's checking whether the price is above or below some certain amount. As prices go up, sales go down, which is just common sense. The tree also checks advertising budget and competitor prices in some of the splits, but those only come into play in certain circumstances.

So, basically, what the tree has learned is that shelf location is the most important thing, then price, and then some other stuff that helps fine-tune the prediction after that.

Most Important Variables:

- Price (100) - this is the most important one overall because it appears in a ton of splits throughout the tree
- Advertising (88) - how much the store spends on local advertising
- Age (72) - the average age of people in the area
- Income (52) - how much money people in the area make
- ShelfLoc (split into Medium and Good) - even though this is the first split, it ranks lower in "importance" because Price appears way more times throughout the tree

Model Over/Under Fitting:

The model is likely not overfitting. The CV RMSE was 2.20 and the test RMSE was 2.04, so the test error is actually a bit lower than what cross-validation predicted. If it was overfitting, the test error would be way higher because the model would have memorized the training data instead of learning real patterns.

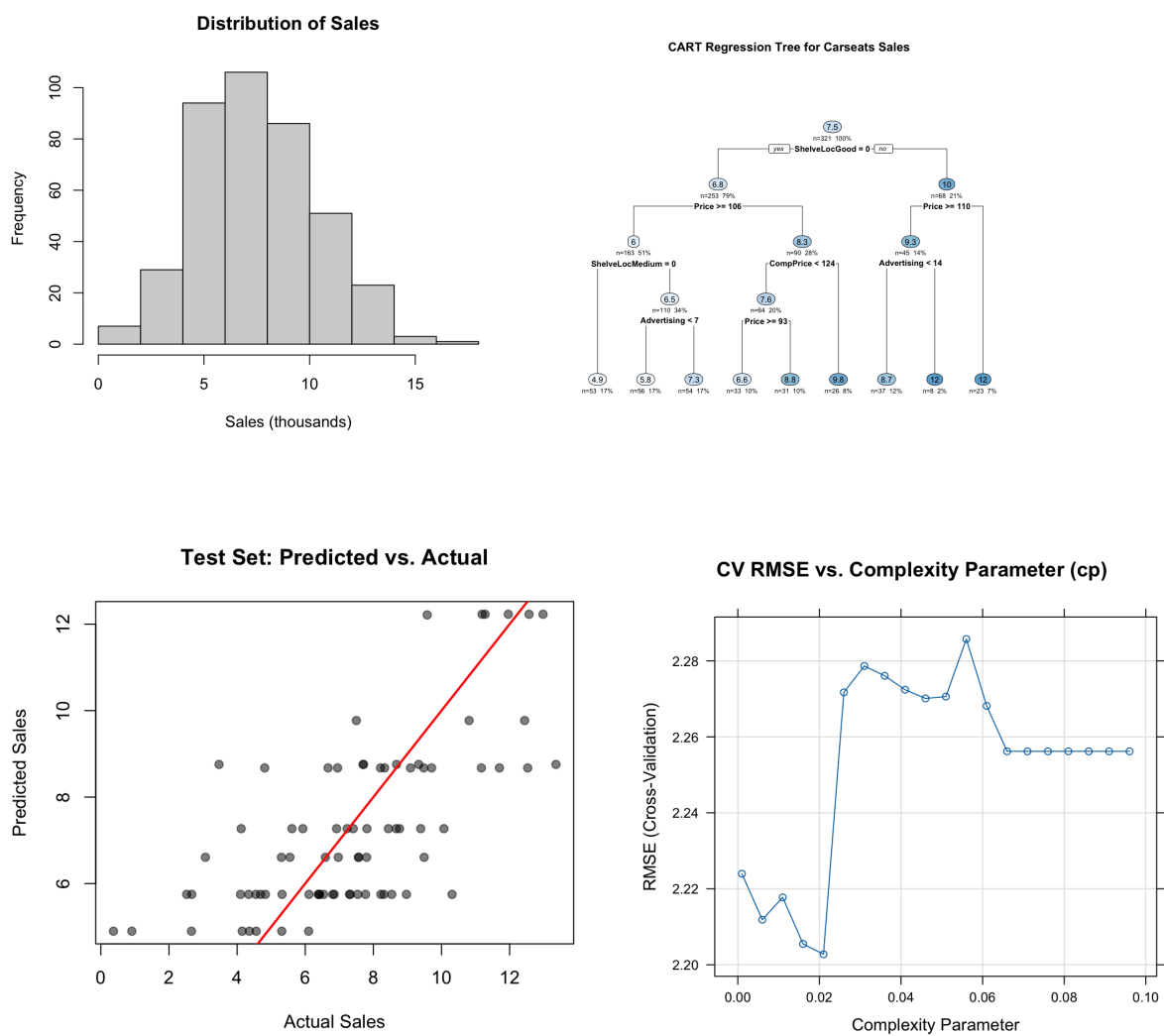
The tuning curve also showed that when I tried really low cp values (which would make super complex trees), the error didn't actually get much better. So making the tree more complicated wouldn't really help.

I think the model is slightly underfitting though. The R-squared is only 0.46, which means the tree is only explaining about half of why sales vary between stores. There's a lot of signal it's

missing. You can also see this in the predicted vs actual plot where the predictions form these horizontal bands because the tree can only output like 8-9 different values (one for each leaf node). So if the true sales is 7.2, the tree might predict 6.6 or 7.3 depending on which bucket the store falls into, but it can't predict 7.2 exactly.

This is probably just a limitation of using a single decision tree. It can only make these rectangular splits and can't capture more complicated relationships. I'm guessing random forests or other methods will do better since they can combine multiple trees or fit the data in different ways.

Plots:



Variable Importance

